1. The B.Tech. (Mechanical Engineering) programme at BIT Mesra consists of 168 credits distributed across various theory and laboratory courses. Each theory course typically carries a 3 - 4 credit load and is divided into five modules, each spanni - 10 lecture hours and defined by specific Course Outcomes (COs).

2. The assessment framework includes Progressive Evaluation (Continuous Assessment - 25% weightage), Mid-Semester Examination (25%), and End-Semester Examination (50%). For theory courses, continuous assessment consists of two quiz (Quiz 1 before the mid-semester examination and Quiz 2 after mid-semester examination), assignments, and classroom interactions.

3. Evaluation Methods: Quizzes (10% each): Quiz 1 is conducted after completion of 1 - 2 modules, while Quiz 2 is held later in the semester. Questions are mapped to COs, and marks obtained are used to assess attainment. Assessment (5%): Assessment marks are awarded through assignments, seminars, surprise tests, attendance, classroom interactions, etc., at the discretion of the individual faculty member. Mid-Semester Examination (25%): Typically consists of five structured question (with sub-parts) covering relevant modules, with each question explicitly marked. Evaluation of Laboratory courses is based on a 60% progressive evaluation and a 40% End Semester exam.

4. The attainment level for each student is calculated by combining the scores from Continuous Assessment, Mid-Semester, and End-Semester Examinations. These attainment levels are categorized on a 3-point scale: Low (≤44.99%), Medium (45 64.99%), High (≥65%).

Since all COs are mapped to Program Outcomes (POs) and Program Specific Outcomes (PSOs), the process ensures that student performance in each course contributes systematically to the attainment of the overall outcome-based B.Tech (Mechan Engineering) programme.

3.2 Evaluation of the Semester End Exam (SEE) Question Paper (10)

The End Semester Examination carries a 50% weightage of the total course evaluation. Question papers are designed to cover the entire course syllabus, incorporating different levels of Bloom's Taxonomy to test knowledge, understanding, applicatio higher-order problem-solving skills. Typically, five main questions (with sub-parts) are included to ensure representation of all modules. Each question is carefully mapped to the respective Course Outcomes (COs), and the maximum marks for each are clearly indicated on the question paper.

§ Students responses are evaluated as per the defined marking scheme, and the marks obtained are recorded in departmental records. These marks are then tabulated (using Excel or equivalent tools) to calculate the percentage attainment of each C through the SEE.

§ The final attainment of each student for every CO is computed by combining scores from Continuous Assessment (Quizzes, Assignments), Mid Semester Examination, and End Semester Examination. The consolidated attainment is categorized on a point scale:

- Low (≤44.99%)
- Medium (45–64.99%)
- High (≥65%)

As all COs are systematically mapped to Program Outcomes (POs) and Program Specific Outcomes (PSOs), this structured evaluation ensures that the B. Tech. (Mechanical Engineering) programme achieves its outcome-based assessment objective effectively.

3.3 Evaluation of Laboratory Work and Workshop (Continuous and SEE) (10)

This structured evaluation ensures that laboratory learning not only develops experimental and analytical skills but also strengthens the attainment of COs, POs, and PSOs of the B. Tech. (Mechanical Engineering) programme.

1. Laboratory and workshop courses in the B. Tech. (Mechanical Engineering) programme carry their own course codes, with a credit weightage typically 50% of the corresponding theory course. While the credits are half, the contact hours are to er adequate hands-on exposure. Assessment of laboratory courses involves multiple components: two quizzes (Quiz 1 before the End Semester Examination and Quiz 2 during/alongside the End Semester Examination), continuous laboratory performan each experiment, viva-voce for every experiment, and timely submission of laboratory records. Marks awarded for each component are mapped to the respective Course Outcomes (COs), ensuring constructive alignment.

2. At the conclusion of the course, students submit their Lab Records to the concerned faculty, which are evaluated along with quiz scores, viva, and practical performance. The attainment of each student for every CO is calculated from the Quiz 1 Record, Viva, and the End Semester Lab Examination. The consolidated attainment levels are then classified on a 3 -point scale:

- Low (≤ 44.99%)
- Medium (45 - 64.99%)

High (≥ 65%)

3.4 Evaluation of Industrial Training/ Internship (Continuous and SEE) (10)

Project Report Evaluation Rubrics :

Performance Indicators	Excellent (Full Marks)	Satisfactory (Mid Marks)	Needs Improvement (Low Marks)	Max Marks	Evaluat
Report Format	Fully adhered (10)	Partially adhered (7)	Not adhered (5)	10	
Clarity & chapter organization	Well defined & clear (10)	Few chapters clear (7)	Not clear (5)	10	
Description of project details	All details described(10)	Few details described (7)	Poorly described (5)	10	
Discussion of results &conclusion	Well discussed (15)	Discussed (12)	Poorly discussed (8)	15	
References	Fully appropriate (5)	Partially appropriate (3)	Inappropriate (2)	5	

Viva-Voce Evaluation Rubrics:

Criteria	Excellent (Full Marks)	Satisfactory (Mid Marks)	Needs Improvement (Low Marks)
Knowledge of Fundamentals	Strong grasp, clear explanations, no major errors (10)	Understands concepts but with minor gaps(7)	Weak understanding, many errors (5)
Application of Knowledge (Problem-Solving &Analysis)	Applies theory effectively to real-world problems (10)	Some application ability, partly correct reasoning (7)	Unable to apply knowledge, incorrect reasoning (5)
Technical Depth (Project/Internship Understanding)	Excellent understanding of objectives, methods, tools, outcomes (10)	Partial understanding, explains some aspects (7)	Poor understanding, unable to explain key aspects (5)
Communication Skills	Fluent explanations, appropriate technical terms, confident answers (8)	Adequate but lacks fluency/clarity, moderate confidence (6)	Weak communication, unclear explanations, low confidence (4)
Confidence & Attitude	Highly confident, enthusiastic, professional(6)	Moderately confident, some hesitation (4)	Nervous, lacks confidence, unprofessional(2)
Overall Impression (Creativity, and Critical Thinking)	Shows originality, creativity, independent thinking (6)	Some originality and critical thinking (4)	Little originality, dependent on others (2)

COs	Description
CO1	Demonstrate the ability to prepare and present a structured technical report with clarity, organization, and proper referencing.
CO2	Apply fundamental engineering knowledge and problem-solving skills to analyze and interpret project/internship-related tasks.
CO3	Exhibit technical depth and comprehensive understanding of the project/internship work, including discussion of results andconclusions.
CO4	Communicate effectively through written and oral presentations, demonstrating confidence and professional attitude.
CO5	Display creativity and critical thinking in project execution

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	0	0	2	1	0	0	1	0	3	1	2	2	0	0
CO2	3	3	2	2	2	1	0	0	0	2	1	3	3	2	0
CO3	3	2	2	3	2	0	0	1	0	2	0	3	3	2	3
CO4	1	0	0	0	0	1	0	2	2	3	0	2	1	0	0
CO5	2	2	3	2	2	0	0	0	1	2	2	3	2	3	2

3.5 Evaluation of Projects (20)

Project evaluation consists of the following components

1. Progressive evaluation-- based on the seminar presentation progress of work is judged and necessary recommendations and suggestions to carry out further work is given to the students by internalexaminer at the mid of final semester.
2. End semester Evaluation- 100 Marks based on the report submitted and seminar presentation on the work done by the students' marks is given by internal examiner as well as external examiner-internalexaminer is guide himself/herself and exte another institute at the end of eighth semester.

s of Project evaluation

l Problem Identification & Analysis

2 Design & Research Application

3 Experimentation & Data Interpretation

4 Project Management, Teamwork & Professional Skills

5 Technical Communication: Report Writing & Presentation

ect report evaluation rubrics

Performance Indicators	**Excellent (Full Marks)**	**Satisfactory (Mid marks)**	**Needs Improvement (Low Marks)**	**Max Marks**
Problem identification and objectives	Fully Adhered(10)	Partially adhered (7)	Not adhered (5)	10
Technical Knowledge and application	Well adhered and Clear (10)	Few Chapters Clear (10)	Not Clear(10)	10
Methodology and Experimentation/Simulation	All details Described(20)	Few Details described(12)	Poorly described (8)	20
Project/management and team work	Well discussed(10)	Discussed(7)	Poorly Discussed(5)	10
Total Marks				50

ı-vocational examination rubrics

Satisfactory (Mid Marks)	Needs improvement (LOW Marks)	Max Marks	COs
Some application ability, partly correct reasoning (12)	Unable to apply knowledge, incorrect reasoning (8)	20	3
Partial understanding, explains some aspects (7)	Poor understanding, unable to explain key aspects (5)	10	5
Adequate but lacks fluency/clarity, moderate confidence (12)	Weak communication, unclear explanations, low confidence (8)	20	1-5
		50	

PO/PSO Attainment matrix

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	3	2	2	1	2	2	1	1	1	1	2	3	2
CO2	3	2	3	2	3	1	2	1	2	1	2	2	3	3
CO3	2	2	2	3	3	1	1	1	1	1	1	2	2	3
CO4	1	1	2	1	1	1	1	2	3	2	3	2	2	2
CO5	1	1	1	1	1	1	1	1	2	3	2	2	1	1

3.6 Evidence of Addressing Sustainable Development Goals (SDG) (10)

Sl. No.	Roll No.	Student Name	Guide Name	Title [Mention if with industry Collaboration]	SDG
1	BTECH/10711/19	Rooturaj Swain	Dr. Mukesh Sharma	Numerical Investigations On The Thermal Behaviour Of A Fully Voided Concentric Channel Of 220 Mwe IPHWR Under Depleting Moderator Condition	Aligned with
2	BTECH/10548/19	Aman Singh	Dr. Mukesh Sharma	Numerical Investigations On The Thermal Behaviour Of A Fully Voided Concentric Channel Of 220 Mwe IPHWR Under Depleting Moderator Condition	Aligned with
3	BTECH/10593/19	Tanmay Gocher	Dr. Mukesh Sharma	Numerical Investigations On The Thermal Behaviour Of A Fully Voided Concentric Channel Of 220 Mwe IPHWR Under Depleting Moderator Condition	Aligned with
4	BTECH/10217/19	Jayant Krishna	Dr. Paritosh Mahata	Design and Analysis of a Cageless Open Differential	Aligned with
5	BTECH/10385/19	Swati Pradhan	Dr. Paritosh Mahata	Design and Analysis of a Cageless Open Differential	Aligned with
6	BTECH/10550/19	Dasari Bhavya	Dr. Paritosh Mahata	Design and Analysis of a Cageless Open Differential	Aligned with
7	BTECH/10534/19	Shubham Sheshank	Dr. Anil Chandra Mahato	Modelling and Analysis of an Automotive Seating Track using Finite Element Analysis	Aligned with
8	BTECH/10610/19	Srinivas Challa	Dr. Anil Chandra Mahato	Modelling and Analysis of an Automotive Seating Track using Finite Element Analysis	Aligned with
9	BTECH/10647/19	Sanskar Zode	Dr. Anil Chandra Mahato	Modelling and Analysis of an Automotive Seating Track using Finite Element Analysis	Aligned with
10	BTECH/10421/19	Shreya Prasad	Saurav Chakraborty	Convective drying of food products	Aligned with
11	BTECH/10453/19	Shubhankar Sinha	Saurav Chakraborty	Convective drying of food products	Aligned with
12	BTECH/10529/19	Fahim Yousuf Choudhury	Saurav Chakraborty	Convective drying of food products	Aligned with
13	BTECH/10548/19	Aman Singh	Dr. Mukesh Sharma	Effect of Moderator Depletion and Investigations on the Thermal Behaviour of 220 Mwe IPHWR Nuclear Channel during LOCA	Aligned with
14	BTECH/10711/19	Rooturaj Swain	Dr. Mukesh Sharma	Effect of Moderator Depletion and Investigations on the Thermal Behaviour of 220 Mwe IPHWR Nuclear Channel during LOCA	Aligned with
15	BTECH/10593/19	Tanmay Gocher	Dr. Mukesh Sharma	Effect of Moderator Depletion and Investigations on the Thermal Behaviour of 220 Mwe IPHWR Nuclear Channel during LOCA	Aligned with
16	BTECH/10627/19	Gourav Kumar	Dr. Mukesh Sharma	Numerical Investigations on low viscous oil-water flow through macro-channel return bend	Aligned with
17	Btech/10493/19	Ritik Raj Sharma	Dr. Mukesh Sharma	Numerical Investigations on low viscous oil-water flow through macro-channel return bend	Aligned with
18	Btech/10559/19	Agraj	Dr. Mukesh Sharma	Numerical Investigations on low viscous oil-water flow through macro-channel return bend	Aligned with
19	BTECH/10529/19	Fahim Yousuf Choudhury	Saurav Chakraborty	Computational Modelling of Convective Drying of Food Products.	Aligned with
20	BTECH/10453/19	Shubhankar Sinha	Saurav Chakraborty	Computational Modelling of Convective Drying of Food Products.	Aligned with
21	BTECH/10421/19	Shreya Prasad	Saurav Chakraborty	Computational Modelling of Convective Drying of Food Products.	Aligned with
22	BTECH/10224/19	Vaddi Srikanth	Abhijith Nag	MODELLING AND OPTIMIZATION OF HARDNESS OF ELECTROLESS NI–B COATING BY USING ANN-GA METHODOLOGY	Aligned with
23	BTECH/10327/19	Bhagawansingh rathor	Abhijith Nag	MODELLING AND OPTIMIZATION OF HARDNESS OF ELECTROLESS NI–B COATING BY USING ANN-GA METHODOLOGY	Aligned with
24	BTECH/10362/19	Adarsh raj	Abhijith Nag	MODELLING AND OPTIMIZATION OF HARDNESS OF ELECTROLESS NI–B COATING BY USING ANN-GA METHODOLOGY	Aligned with
25	BTECH/10416/19	Nishit Mishra	Praveen James Sanga	COMPREHENSIVE STUDY ON THE MECHANICAL AND DYNAMIC PROPERTIES OF AIRLESS TIRE DESIGNS	Aligned with
26	BTECH/10504/19	Kumar Prasoon	Praveen James Sanga	COMPREHENSIVE STUDY ON THE MECHANICAL AND DYNAMIC PROPERTIES OF AIRLESS TIRE DESIGNS	Aligned with
27	BTECH/10531/19	Aurajeet Mahapatra	Praveen James Sanga	COMPREHENSIVE STUDY ON THE MECHANICAL AND DYNAMIC PROPERTIES OF AIRLESS TIRE DESIGNS	Aligned with
28	BTECH/10611/19	Prakhar Shah	Dr. Arun Kumar Kadian	Design and Analysis of roll cage and braking system of an All-Terrain Vehicle.	Aligned with
29	BTECH/10135/19	Sahil Kumar Pandey	Dr. Arun Kumar Kadian	Design and Analysis of roll cage and braking system of an All-Terrain Vehicle.	Aligned with
30	BTECH/10108/19	Vivek Kumar	Dr. Arun Kumar Kadian	Design and Analysis of roll cage and braking system of an All-Terrain Vehicle.	Aligned with
31	BTECH.10630.19	Samyak Kumar	Mukesh Sharma	To develop a brake system that is capable of locking four wheels altogether at a max deceleration of 1.5g	Aligned with
32	BTECH.10630.19	Pranshu Singh	Mukesh Sharma	To develop a brake system that is capable of locking four wheels altogether at a max deceleration of 1.5g	Aligned with
33	BTECH/10048/19	Biswarup Roy	Dr. Om Prakash	Mathematical Modelling and Simulation of Helical Coil Based Heat Exchanger for Concentrated Solar Power Plant	Aligned with
34	BTECH/10276/19	Vivek Kumar	Dr. Om Prakash	Mathematical Modelling and Simulation of Helical Coil Based Heat Exchanger for Concentrated Solar Power Plant	Aligned with
35	BTECH/10342/19	D Abhaya Kumar	Dr. Om Prakash	Mathematical Modelling and Simulation of Helical Coil Based Heat Exchanger for Concentrated Solar Power Plant	Aligned with
36	BTECH/10175/19	Shubham Mishra	Dr. S.K. Dhiman	DESIGN AND DEVELOPMENT OF A PORTABLE AIR-COOLING SYSTEM USING WATER WITHOUT ANY EVAPORATION LOSSES	Aligned with
37	BTECH/10062/19	Abhijeet Pranav Mishra	Dr. S.K. Dhiman	DESIGN AND DEVELOPMENT OF A PORTABLE AIR-COOLING SYSTEM USING WATER WITHOUT ANY EVAPORATION LOSSES	Aligned with
38	BTECH/10173/19	Ayush Kumar	Dr. S.K. Dhiman	DESIGN AND DEVELOPMENT OF A PORTABLE AIR-COOLING SYSTEM USING WATER WITHOUT ANY EVAPORATION LOSSES	Aligned with
39	BTECH/10430/19	Aman Srivastava	Dr. Praveen Mishra	Design and analysis of gearbox for an all-terrain vehicle	Aligned with
40	BTECH/10637/19	Akshat Srivastava	Dr. Praveen Mishra	Design and analysis of gearbox for an all-terrain vehicle	Aligned with
41	BTECH/10684/19	Aditya Kumar Panda	Dr. Praveen Mishra	Design and analysis of gearbox for an all-terrain vehicle	Aligned with
42	BTECH/10158/19	Anik Sarkar	Saurav Chakraborty	STUDY OF INVERSE HEAT TRANSFER TECHNIQUES IN CONTINUOUS CASTING	Aligned with

43BTECH/10357/19	Rishav Deo Raj	Saurav Chakraborty	STUDY OF INVERSE HEAT TRANSFER TECHNIQUES IN CONTINUOUS CASTING	Aligned with
44BTECH/10431/19	Ayush Jacob kujur	Om prakash pandey	DESIGN, FABRICATION AND COMPARISON OF CUBOIDAL AND CYLINDRICAL HYDRAULIC TURBINE	Aligned with
45BTECH/10734/19	Rohit Roshan Topno	Om prakash pandey	DESIGN, FABRICATION AND COMPARISON OF CUBOIDAL AND CYLINDRICAL HYDRAULIC TURBINE	Aligned with
46BTECH/10537/19	Kamal Gautam	Om prakash pandey	DESIGN, FABRICATION AND COMPARISON OF CUBOIDAL AND CYLINDRICAL HYDRAULIC TURBINE	Aligned with
47BTECH/10723/19	MUDAVATH RAVI	Dr.OM PRAKASH	MATHEMATICAL MODELLING AND SIMULATION OF HELICAL COIL BASED HEAT EXCHANGER FOR CONCENTRATED SOLAR POWER PLANT	Aligned with
48BTECH/10048/19	BISWARUP ROY	Dr.OM PRAKASH	MATHEMATICAL MODELLING AND SIMULATION OF HELICAL COIL BASED HEAT EXCHANGER FOR CONCENTRATED SOLAR POWER PLANT	Aligned with
49BTECH/10276/19	VIVEK KUMAR	Dr.OM PRAKASH	MATHEMATICAL MODELLING AND SIMULATION OF HELICAL COIL BASED HEAT EXCHANGER FOR CONCENTRATED SOLAR POWER PLANT	Aligned with
50BTECH/10342/19	D ABHAYA KUMAR	Dr.OM PRAKASH	MATHEMATICAL MODELLING AND SIMULATION OF HELICAL COIL BASED HEAT EXCHANGER FOR CONCENTRATED SOLAR POWER PLANT	Aligned with
51BTECH/10093/19	Sohini Shah	Dr. Rajeev Kumar	DESIGN, FABRICATION AND PERFORMANCE ANALYSIS OF A COMPRESSED AIR ENGINE THROUGH GREEN ENERGY RESOURCES	Aligned with
52BTECH/10269/19	Ronit Kumar	Dr. Rajeev Kumar	DESIGN, FABRICATION AND PERFORMANCE ANALYSIS OF A COMPRESSED AIR ENGINE THROUGH GREEN ENERGY RESOURCES	Aligned with
53BTECH/10271/19	Neeraj Narang	Dr. Rajeev Kumar	DESIGN, FABRICATION AND PERFORMANCE ANALYSIS OF A COMPRESSED AIR ENGINE THROUGH GREEN ENERGY RESOURCES	Aligned with
54BTECH/10194/19	Aklavya Anurag	Dr. Om Prakash	Experimental and Computational Study of Plate Steam Condenser Heat Transfer Characteristics	Aligned with
55BTECH/10203/19	Akash Raj	Dr. Om Prakash	Experimental and Computational Study of Plate Steam Condenser Heat Transfer Characteristics	Aligned with
56BTECH/10417/19	Pranav Prakash	Dr. Om Prakash	Experimental and Computational Study of Plate Steam Condenser Heat Transfer Characteristics	Aligned with
57BTECH/10598/19	Amod Surin	Arkadeb Mukhopadhyay	DEVELOPMENT OF BI- LAYER ELECTROLESS Ni-Sn-Cu-B COATING AND INVESTIGATION OF ITS TRIBOLOGICAL BEHAVIOUR	Aligned with
58BTECH/10025/19	Arpit Nischal	Arkadeb Mukhopadhyay	DEVELOPMENT OF BI- LAYER ELECTROLESS Ni-Sn-Cu-B COATING AND INVESTIGATION OF ITS TRIBOLOGICAL BEHAVIOUR	Aligned with
59BTECH/10148/19	Vivek Gari	Arkadeb Mukhopadhyay	DEVELOPMENT OF BI- LAYER ELECTROLESS Ni-Sn-Cu-B COATING AND INVESTIGATION OF ITS TRIBOLOGICAL BEHAVIOUR	Aligned with
60BTECH/10364/19	Abhishek Ranjan	Dr. Mukesh Sharma	Analysis of phase distribution during kerosene-water flow through inverted U bend	Aligned with
61BTECH/10638/19	Aditya Abhay Jain	Dr. Mukesh Sharma	Analysis of phase distribution during kerosene-water flow through inverted U bend	Aligned with
62BTECH/10409/19	Varad Gangadhar Gattuwar	Dr. Mukesh Sharma	Analysis of phase distribution during kerosene-water flow through inverted U bend	Aligned with
63BTECH/10630/19	Samyak Kumar	Mukesh Sharma	To develop a brake system that is capable of locking four wheels altogether at a max deceleration of 1.5g	Aligned with
64BTECH/10475/19	Pranshu Singh	Mukesh Sharma	To develop a brake system that is capable of locking four wheels altogether at a max deceleration of 1.5g	Aligned with
65BTECH/10296/19	Nilesh Ranjan	Prof. Praveen James Sanga	HEAT TRANSFER ENHNACEMENT OF TURBULENT FORCED NANOFLUID FLOW IN A MICROCHANNEL USING SECTOR SHAPED RIB	Aligned with
66BTECH/10304/19	Nitish Kumar	Prof. Praveen James Sanga	HEAT TRANSFER ENHNACEMENT OF TURBULENT FORCED NANOFLUID FLOW IN A MICROCHANNEL USING SECTOR SHAPED RIB	Aligned with
67BTECH/10310/19	Suman Shekhar	Prof. Praveen James Sanga	HEAT TRANSFER ENHNACEMENT OF TURBULENT FORCED NANOFLUID FLOW IN A MICROCHANNEL USING SECTOR SHAPED RIB	Aligned with
68BTECH/10147/19	SANGAM BHARTI	Dr Nirmal Kumar	ANALYSIS OF FATIGUE CRACK INITIATION IN CYCLIC MICROPLASTICITY REGIMES THROUGH A NON-INVASIVE TECHNIQUE	Aligned wi developm
69BTECH/10354/19	SONU SAHA	Dr Nirmal Kumar	ANALYSIS OF FATIGUE CRACK INITIATION IN CYCLIC MICROPLASTICITY REGIMES THROUGH A NON-INVASIVE TECHNIQUE	Aligned wi developm
70BTECH/10594/19	HARSH RANJAN	Dr Nirmal Kumar	ANALYSIS OF FATIGUE CRACK INITIATION IN CYCLIC MICROPLASTICITY REGIMES THROUGH A NON-INVASIVE TECHNIQUE	Aligned wi developm
71BTECH/10675/19	Abhishek	Dr. Rajeev Kumar	INVESTIGATIONS ON MASS ENERGY BALANCE ANALYSIS OF A BIOGAS GENERATION PLANT	Aligned with
72BTECH/10447/19	Shreya Shukla	Dr. Rajeev Kumar	INVESTIGATIONS ON MASS ENERGY BALANCE ANALYSIS OF A BIOGAS GENERATION PLANT	Aligned with
73BTECH/10033/19	Kumari Kritika	Dr. Rajeev Kumar	INVESTIGATIONS ON MASS ENERGY BALANCE ANALYSIS OF A BIOGAS GENERATION PLANT	Aligned with
74Btech/10593/19	Tanmay Gocher	Mukesh Sharma	Effect of moderator depletion and investigations on the thermal behaviour of 220 Mwe IPHWR nuclear channel during LOCA	Aligned with
75Btech/10548/19	Rooturaj Swain	Mukesh Sharma	Effect of moderator depletion and investigations on the thermal behaviour of 220 Mwe IPHWR nuclear channel during LOCA	Aligned with
76Btech/10711/19	Aman Singh	Mukesh Sharma	Effect of moderator depletion and investigations on the thermal behaviour of 220 Mwe IPHWR nuclear channel during LOCA	Aligned with
77BTECH/10147/19	SANGAM BHARTI	Dr NIRMAL KUMAR	ANALYSIS OF FATIGUE CRACK INITIATION IN CYCLIC MICROPLASTICITY REGIMES THROUGH A NON-INVASIVE TECHNIQUE	Aligned with
78BTECH/10354/19	SONU SAHA	Dr NIRMAL KUMAR	ANALYSIS OF FATIGUE CRACK INITIATION IN CYCLIC MICROPLASTICITY REGIMES THROUGH A NON-INVASIVE TECHNIQUE	Aligned with
79BTECH/10594/19	Harsh Ranjan	Dr NIRMAL KUMAR	ANALYSIS OF FATIGUE CRACK INITIATION IN CYCLIC MICROPLASTICITY REGIMES THROUGH A NON-INVASIVE TECHNIQUE	Aligned with
80BTECH/10392/19	Rakshit Jain	Dr. Richa Pandey	Design and analysis of Vehicle Dynamics system for an all terrain vehicle	Aligned with
81BTECH/10425/19	Shubhankar	Dr. Richa Pandey	Design and analysis of Vehicle Dynamics system for an all terrain vehicle	Aligned with
82BTECH/10607/19	Chetan Sinha	Dr. Richa Pandey	Design and analysis of Vehicle Dynamics system for an all terrain vehicle	Aligned with
83BTECH/10377/19	Ankit Kumar	Dr. Abhijit Nag	Optimization of dimension in a pressure vessel cylinder	Aligned with

84	BTECH/10370/19	Harsh Barnwal	Dr. Abhijit Nag	Optimization of dimension in a pressure vessel cylinder	Aligned with
85	BTECH/10239/19	Deva Ajay Kumar	Dr. Abhijit Nag	Optimization of dimension in a pressure vessel cylinder	Aligned with
86	BTECH/10235/19	K Jayanth Kumar	Om Prakash	CFD modelling and experimentation of steam turbine used in concentrated solar power plant	Aligned with
87	BTECH/10281/19	Kuthadi Harshavardhan	Om Prakash	CFD modelling and experimentation of steam turbine used in concentrated solar power plant	Aligned with
88	BTECH/10275/19	P Vishnu	Om Prakash	CFD modelling and experimentation of steam turbine used in concentrated solar power plant	Aligned with
89	Btech/10671/19	Gautam Gupta	Richa Pandey	Design and development of automatic cradle	Aligned with
90	Btech/10687/19	Anish oraon	Richa Pandey	Design and development of automatic cradle	Aligned with
91	Btech/10659/19	Madhav Agarwal	Richa Pandey	Design and development of automatic cradle	Aligned with
92	BTECH/10138/19	VISHAL SARDAR	Dr. Om Prakash Pandey	FABRICATION AND EXPERIMENTAL ANALYSIS OF GEAR EMBEDDED LOW HEAD CROSS FLOW TURBINE	Aligned with
93	BTECH/10165/19	ARJUN ORAON	Dr. Om Prakash Pandey	FABRICATION AND EXPERIMENTAL ANALYSIS OF GEAR EMBEDDED LOW HEAD CROSS FLOW TURBINE	Aligned with
94	BTECH/10337/19	CHANDAN MARDI	Dr. Om Prakash Pandey	FABRICATION AND EXPERIMENTAL ANALYSIS OF GEAR EMBEDDED LOW HEAD CROSS FLOW TURBINE	Aligned with
95	BTECH/10093/19	Sohini Shah	Dr. Rajeev Kumar	DESIGN, FABRICATION AND PERFORMANCE ANALYSIS OF A COMPRESSED AIR ENGINE THROUGH GREEN ENERGY RESOURCES	Aligned with
96	BTECH/10269/19	Ronit Kumar	Dr. Rajeev Kumar	DESIGN, FABRICATION AND PERFORMANCE ANALYSIS OF A COMPRESSED AIR ENGINE THROUGH GREEN ENERGY RESOURCES	Aligned with
97	BTECH/10271/19	Neeraj Narang	Dr. Rajeev Kumar	DESIGN, FABRICATION AND PERFORMANCE ANALYSIS OF A COMPRESSED AIR ENGINE THROUGH GREEN ENERGY RESOURCES	Aligned with
98	BTECH/10479/19	Anuj Agrawal	Dr. Richa Pandey	Design and development of shredder machine for efficient recycling of plastic in India	Aligned with
99	BTECH/10554/19	Somya Garg	Dr. Richa Pandey	Design and development of shredder machine for efficient recycling of plastic in India	Aligned with
100	BTECH/10666/19	Prateek Shankar Prasad	Dr. Richa Pandey	Design and development of shredder machine for efficient recycling of plastic in India	Aligned with
101	BTECH/10670/19	Adrash Raj	Dr. Richa Pandey	Design and development of shredder machine for efficient recycling of plastic in India	Aligned with
102	BTECH/10601/19	Anuj Pandey	Dr. Arun Kumar Kadian	DESIGNING AND FINITE ELEMENT ANALYSIS OF SUSPENSION AND STEERING GEOMETRY OF AN ALL TERRAIN VEHICLE	Aligned with
103	BTECH/10603/19	Shritam Sain	Dr. Arun Kumar Kadian	DESIGNING AND FINITE ELEMENT ANALYSIS OF SUSPENSION AND STEERING GEOMETRY OF AN ALL TERRAIN VEHICLE	Aligned with
104	BTECH/10653/19	Aniruddh K Singh	Dr. Arun Kumar Kadian	DESIGNING AND FINITE ELEMENT ANALYSIS OF SUSPENSION AND STEERING GEOMETRY OF AN ALL TERRAIN VEHICLE	Aligned with
105	BTECH/10034/19	Siddhartha Shekhar Mishra	Dr. Praveen James Sanga	NUMERICAL INVESTIGATION OF HEAT TRANSFER PERFORMANCE OF A DUCT IN A RADIATOR USING NANOFLUID	Aligned with
106	BTECH/10122/19	Rahul Sinha	Dr. Praveen James Sanga	NUMERICAL INVESTIGATION OF HEAT TRANSFER PERFORMANCE OF A DUCT IN A RADIATOR USING NANOFLUID	Aligned with
107	BTECH/10495/19	Ashutosh Bhadauri	Dr. Praveen James Sanga	NUMERICAL INVESTIGATION OF HEAT TRANSFER PERFORMANCE OF A DUCT IN A RADIATOR USING NANOFLUID	Aligned with
108	BTECH/10706/19	Lokesh Gupta	Dr. Arkadeb Mukhopadhyay	TRIBOLOGICAL STUDY OF BI-LAYER ELECTROLESS Ni-B COATING ON AISI 1040 STEEL	Aligned with
109	BTECH/10617/19	Ritik Sharma	Dr. Arkadeb Mukhopadhyay	TRIBOLOGICAL STUDY OF BI-LAYER ELECTROLESS Ni-B COATING ON AISI 1040 STEEL	Aligned with
110	BTECH/10679/19	Siddharth Jain	Dr. Arkadeb Mukhopadhyay	TRIBOLOGICAL STUDY OF BI-LAYER ELECTROLESS Ni-B COATING ON AISI 1040 STEEL	Aligned with
111	Btech/10390/19	Sumedh Dintakurthy	Anil chandra Mahato	DESIGN AND ANALYSIS OF VARIABLE INERTIA FLYWHEEL	Aligned with
112	Btech/10701/19	Peala yeswanth	Anil chandra Mahato	DESIGN AND ANALYSIS OF VARIABLE INERTIA FLYWHEEL	Aligned with
113	Btech/10563/19	Gurram Sunny Aroma	Anil chandra Mahato	DESIGN AND ANALYSIS OF VARIABLE INERTIA FLYWHEEL	Aligned with
114	BTECH/10036/19	Shashank Kumar Singh	Dr. Paritosh Mahata	Design, Development and Analysis of the Convertible Stairs	Aligned with
115	BTECH/10064/19	Rahul Roshan	Dr. Paritosh Mahata	Design, Development and Analysis of the Convertible Stairs	Aligned with
116	BTECH/10280/19	Roshan Manish Agarwa	Dr. Paritosh Mahata	Design, Development and Analysis of the Convertible Stairs	Aligned with
117	BTECH/10455/19	Aditya Kumar	Dr. Om Prakash	Energy and Exergy analysis of innovative steam condenser	Aligned with
118	BTECH/10724/19	Vijay Kumar Das	Dr. Om Prakash	Energy and Exergy analysis of innovative steam condenser	Aligned with
119	BTECH/10521/19	Ankit Kumar	Dr. Om Prakash	Energy and Exergy analysis of innovative steam condenser	Aligned with
120	BTECH/10467/19	Devesh Kujur	Dr. Richa Pandey	A Study of Effect of Heat Treatment on Mechanical Properties of 3D Printed Polylactic Acid	Aligned with
121	BTECH/10159/19	Runel Kerketta	Dr. Richa Pandey	A Study of Effect of Heat Treatment on Mechanical Properties of 3D Printed Polylactic Acid	Aligned with
122	BTECH/10176/19	Rishabh Toppo	Dr. Richa Pandey	A Study of Effect of Heat Treatment on Mechanical Properties of 3D Printed Polylactic Acid	Aligned with
123	BTECH/10306/19	DIVYANSHU	DR. OM PRAKASH	DESIGN AND DEVELOPMENT OF THERMAL OIL BASED PARABOLIC TROUGH COLLECTOR	Aligned with
124	BTECH/10743/19	VIPIN KUMAR	DR. OM PRAKASH	DESIGN AND DEVELOPMENT OF THERMAL OIL BASED PARABOLIC TROUGH COLLECTOR	Aligned with
125	BTECH/10130/19	SUBHAM PASHI	DR. OM PRAKASH	DESIGN AND DEVELOPMENT OF THERMAL OIL BASED PARABOLIC TROUGH COLLECTOR	Aligned with
126	BTECH/10059/19	Siddhant Pradhan	Dr. Om Prakash	Mathematical Modelling and Simulation of Helical Coil Based Heat Exchanger for Concentrated Solar Power Plant	Aligned with
127	BTECH/10324/19	Harshit Kejriwal	Dr. Om Prakash	Mathematical Modelling and Simulation of Helical Coil Based Heat Exchanger for Concentrated Solar Power Plant	Aligned with
128	BTECH/10411/19	Devlin Sarkar	Dr. Om Prakash	Mathematical Modelling and Simulation of Helical Coil Based Heat Exchanger for Concentrated Solar Power Plant	Aligned with
129	BTECH/10568/19	ADITYA SINGH JARIAL	Dr. Sujeet Kumar Mishra	Designing of IoT framework for machine digitalization	Aligned with
130	BTECH/10538/19	EKANSH ZHARIYA	Dr. Sujeet Kumar Mishra	Designing of IoT framework for machine digitalization	Aligned with
131	BTECH/10587/19	SHUBHAM KUMAR	Dr. Sujeet Kumar Mishra	Designing of IoT framework for machine digitalization	Aligned with
132	BTECH/10678/19	Dishant Katdare	Sujeet Kumar Mishra	Designing of a torsionally stiff race car chassis	Aligned with
133	BTECH/10703/19	Shashwat Chaturvedi	Sujeet Kumar Mishra	Designing of a torsionally stiff race car chassis	Aligned with

134	BTECH/10681/19	Akshay Tando	Sujeet Kumar Mishra	Designing of a torsionally stiff race car chassis	Aligned with
135	Btech/10473/19	Rishav Mishra	Dr. Richa Pandey	Fabrication of foot orthotic insole by 3D printing	Aligned with
136	Btech/10686/19	Rishabh Thakur	Dr. Richa Pandey	Fabrication of foot orthotic insole by 3D printing	Aligned with
137	Btech/10287/19	Priya Agarwal	Dr. Richa Pandey	Fabrication of foot orthotic insole by 3D printing	Aligned with

3.7 Attainment of Course Outcomes (25)

3.7.1. Describe the Assessment Tools and Processes Used to Gather the Data for the Evaluation of Course Outcome (5)

I. EXAMINATIONS FOR THE UNDER-GRADUATE (UG) PROGRAMME

1. **THEORY COURSES**: The total of 100 marks are distributed as under:

   a) **Semester End Examination [50%]**

   The Question Paper will have 05 questions with one question from each of the five modules of the syllabus, and there will be no choice. All 05 questions need to be answered to ensure compliance of course outcome as required by graduate attributes.

   b) **Continuous Internal Assessment [50%]**

   The Continuous Internal Assessment shall consist of:

   i) Mid-Semester Examination [25%]

   Mid-semester examination will be conducted by the Examination Section of the Institute. The Question Paper shall not carry any choice. There shall be five (05) questions, each carrying 5 marks (*5×5= 25 marks*).

   ii) Two Quizzes [20% *(2 X 10%)***]**

   There shall be two quizzes of 10 marks each. First Quiz will be held before the mid-semester examination and the Second Quiz before the Semester-End examination. The two quizzes will be conducted by the department offering that course.

   iii) Teacher's Assessment [5%]

   This assessment shall be decided by the teacher(s) conducting that course. This may consist of assignment(s) / attendance / performance / participation in class/ viva. However, the chosen assessment tool shall be announced in the class by the teacher(s) before its commencement and shall be the same for all sections of that course and for the same course at all Centres where this course is being offered.

Common courses taught for multiple sections of the same paper, Quiz (es) across all sections shall be uniform i.e., same Question Paper to be used for the quizzes and the quizzes to be conducted on the same day and at the same time. For common papers, the department shall take the service of one common coordinator who shall be responsible for setting up the Question Paper in consultation with the course teacher(s). Coordinators shall be appointed by the respective Head of the Department/Directors of the Centres as the case may be. If a course is taught by more than one teacher, there shall be one common coordinator for the course.

2. **LABORATORY COURSES: [100 Marks** (*Distributed as under*)**]**:

a) **Continuous Internal Assessment** **[60%** *(Distribution is as follows)*]

i)	Day-to-day performance and Lab files	[30%]
ii)	Quiz(es)	[10%]
iii)	Viva	[20%]
	Subtotal of 2.(a)	**[60%]**

Birla Institute of Technology Mesra Ranchi

b) **Semester-End Examination** **[40%** *(Distribution is as follows)*]

i)	Examination Experiment Performance	[30%]
ii)	Quiz	[10%]
	Subtotal of 2.(b)	**[40%]**
	Grand Total 2.(a)+(b)	**[100%]**

3. **UG PROJECT: The total of 100 marks are distributed as under:**

Project work may be carried out preferably independently or by a group of students not exceeding four (4). More than four students in a group shall be discouraged. Students opting for an Industry Internship in lieu of project work shall do it preferably independently or as deemed by the concerned industry. Project work done outside Institution shall be from Institutions with current Engineering/Science/Management/ University NIRF ranking within 100. In the case of Pharmacy/Architecture students, the same shall be from Institutions with current Pharmacy/Architecture NIRF ranking within 50. In any case, there shall be an academic guide from the parent Institution. Students doing industry internships or research projects in an outside Organization/Institute/University shall produce an internship/project completion certificate from the concerned Organisation/Institute/University to the Head of the Department before the final examination.

Project evaluation shall be as follows:

a) External Examiner	[50%]
b) Project Guide	[25%]
c) Departmental Project Evaluation Committee (DPEC)	[25%]
TOTAL	**[100%]**

The Departmental Project Evaluation Committee (DPEC) shall be constituted by the Head of the Department with inputs from and approval of Dean (UGS). The Guide of the candidate shall not participate in the evaluation process by the DPEC. The panel of external examiners to be provided by the Head of the Department shall be from Institutions with current Engineering/Science/ Management/University NIRF ranking within 100.

For Pharmacy and Architecture projects, the panel of external examiners to be provided by the Head of the Department shall be from Institutions with current Pharmacy/Architecture NIRF ranking within 50.
A plagiarism check is mandatory for the evaluation of all the Dissertations.

4. **Guidelines for Summer Training of the Students of UG Sixth Semester**
The Undergraduate students of Sixth Semester under CBCS Course will have to undergo a Summer Training Programme in a reputed Organization, which shall be a part of the Course Curriculum for the award of degree.

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Further, the student will have to prepare a Report, duly certified by the Summer Training Organization soon after the completion of the Summer Training. The Head of the Department shall constitute an Evaluation Committee to conduct the Viva-Voce of these students. The Viva-Voce examination shall be conducted after the completion of Summer Training.

The break-up of marks shall be as under:

a) Progressive evaluation (50%) : Based on Summer Training Report
b) End-semester Evaluation (50%) : Based on Viva-Voce conducted in the Department by a Committee.

3.7.2 Record the Attainment of Course Outcomes of all Courses with Respect to Set Attainment Levels (20)

Steps for Calculating CO, PO/PSO and PEO Attainment

1. Map the Questions to respective COs. Note that one question can map to multiple COs with different weights.
   For eg. Q1 (a) → CO1 (60%), CO2 (40%) → Different weightage
   Q1 (b) →CO1 (50%), CO2 (50%) → Equal weightage
   Map for all forms of assessments.
2. For each student, calculate the following:
   a. For each assessment type, compute the weighted sum of marks obtained in questions mapped to the respective COs and the Total marks those questions carry.
   For eg. For Mid Semester Exam

$$Marks_{CO_x} = \sum_{i=1}^{n} M_i * W_i$$

$$Total_{CO_x} = \sum_{i=1}^{n} FM_i * W_i$$

x is the number of COs. $x = 1, 2, 3, 4, 5, \ldots$
M_i is the marks obtained and FM_i is the full mark for the ith question mapped to CO_x and n is the total number of questions/assessments.
W_i is the weightage of the question i for CO_x. It can be any % value in the range [0, 100]

Questions mapping to CO's											
Assessments (FM)	**Questions**	**Mapped CO's**									
Quiz 1 (10)	Q1, Q2	CO1, CO2									
Quiz 2 (10)	Q1, Q2	CO3, CO4									
Assignment (5)	Q1	CO4, CO5									
Mid Sem (25)	Q1, Q2, Q3, Q4, Q5	1(a)	1(b)	2(a)	2(b)	3(a)	3(b)	4(a)	4(b)	5(a)	5(b)
		CO1	CO1	CO1	CO1	CO2	CO2	CO2	CO2	CO3	CO3
End Sem (50)	Q1, Q2, Q3, Q4, Q5	1(a)	1(b)	2(a)	2(b)	3(a)	3(b)	4(a)	4(b)	5(a)	5(b)
		CO1	CO1	CO2	CO2	CO3	CO3	CO4	CO4	CO5	CO5

Quiz 1 mapped to CO1 (50%) & CO2 (50%)

$Total_{CO_1} = \sum_{i=1}^{3} FM_i * W_i$
$= 10 * \frac{50}{100}$
$= 5$

$Marks_{CO_1} = \sum_{i=1}^{3} M_i * W_i$
$= 5 * \frac{50}{100}$
$= 2.5$

			Internal (25%)			CO1	
	Quiz1(10)		Quiz2(10)		Assignment (5)	$Total_{CO_1}$	$Marks_{CO_1}$
Full Marks	**5**	**5**	**5**	**5**	**5**		
Student 1	3	2	2	4	4	5	2.5
Student 2	2	5	5	3	5	5	3.5
Student 3	1	4	4	2	4	5	2.5
Student 4	4	4	4	4	5	5	4
Student 5	3	5	5	4	5	5	4
Student 6	2	2	2	3	5	5	2
Student 7	5	4	3	5	5	5	4.5
Student 8	3	5	4	2	5	5	4
Student 9	3	0	1	3	4	5	1.5
Student 10	1	4	4	3	5	5	2.5

b. For each assessment type, apply the weightage of the assessment and again calculate the marks obtained for each CO and the Total marks (This step is required only when the total marks of all assessments is not in the ratio prescribe by BIT i.e Internal-25%, Mid Sem-25% and End Sem-50%).

$$Weighted_Total_{CO_x} = Total_{CO_x} * AW\%$$

$$Weighted_Marks_{CO_x} = \frac{Marks_{CO_x}}{Total_{CO_x}} * Weighted_Total_{CO_x}$$

Where AW is the weightage, the assessment carries.

$Weighted_Total_{CO_1} = Total_{CO_1} * AW\%$

$= 5 * \frac{25}{100}$

$= 1.25$

$Weighted_Marks_{CO_1} = \frac{Marks_{CO_1}}{Total_{CO_1}} * Weighted_Total_{CO_1}$

$= \frac{2.5}{5} * 1.25$

$= 0.63$

			Internal (25%)			CO1		CO1	
	Quiz1 (10)		Quiz2 (10)		Assignment (5)	$Total_{CO_1}$	$Marks_{CO_1}$	$Weighted_Total_{CO_1}$	$Weighted_Marks_{CO_1}$
Full Marks	**5**	**5**	**5**	**5**	**5**				
Student 1	3	2	2	4	4	5	2.5	1.25	0.63
Student 2	2	5	5	3	5	5	3.5	1.25	0.88
Student 3	1	4	4	2	4	5	2.5	1.25	0.63
Student 4	4	4	4	4	5	5	4	1.25	1.00
Student 5	3	5	5	4	5	5	4	1.25	1.00
Student 6	2	2	2	3	5	5	2	1.25	0.50
Student 7	5	4	3	5	5	5	4.5	1.25	1.13
Student 8	3	5	4	2	5	5	4	1.25	1.00
Student 9	3	0	1	3	4	5	1.5	1.25	0.38
Student 10	1	4	4	3	5	5	2.5	1.25	0.63

c. For each assessment type, calculate Attainment_Percentage, Attainment_Score and Attainment_Level based on the predefined rubrics, for each CO.

S. No	Evaluator	Score	Grading Scale Min % Range	Grading Scale Max % Range
1	High	3	65	100
2	Medium	2	45	64.99
3	Low	1	0	44.99

$$\text{Attainment_Percentage}_{CO_x} = \frac{Weighted_Marks_{CO_x}}{Weighted_Total_{CO_x}} * 100$$

$$\text{Attainment_Score}_{CO_x} = \begin{cases} 1, if\ 44.9 > Attainment_Percentage > 0 \\ 2, if\ 45 > Attainment_Percentage > 64.99 \\ 3, if\ 65 > Attainment_Percentage > 100 \end{cases}$$

$$\text{Attainment_Level}_{CO_x} = \begin{cases} LOW, if\ Attainment_Score = 1 \\ MEDIUM, if\ \ if\ Attainment_Score = 2 \\ HIGH, if\ \ if\ Attainment_Score = 3 \end{cases}$$

$$\text{Attainment_Percentage}_{CO_1} = \frac{Weighte_Marks_{CO_1}}{Weighted_Total_{CO_1}} * 100 = \frac{0.63}{1.25} * 100$$

$$\text{Attainment_Score/Level}_{CO_1} = \begin{cases} 1/LOW, if\ 44.9 > Attainment_Percentage > 0 \\ \mathbf{2/MEDIUM, if\ 45 > Attainment_Percentage > 64.99} \\ 3/HIGH, if\ 65 > Attainment_Percentage > 100 \end{cases}$$

	CO1		
	Internal		
	Attainment_ Percentage	Attainment_ Score	Attainment_ Level
Student 1	50	2	MEDIUM
Student 2	70	3	HIGH
Student 3	50	2	MEDIUM
Student 4	80	3	HIGH
Student 5	80	3	HIGH
Student 6	40	1	LOW
Student 7	90	3	HIGH
Student 8	80	3	HIGH
Student 9	30	1	LOW
Student 10	50	2	MEDIUM

Calculate this for all the other assessments like Mid Sem and End Sem, before proceeding to the next Step

3. Calculate the Direct_Assessment_Percentage and Direct_Assessment_Score over the total number of students in the course, over all the assessment types. Check if the target is achieved or not.

$$\text{Direct_Assessment_Percentage}_{CO_x} = \sum_{i=1}^{M} \sum_{j=1}^{N} \frac{Assessment_Percentage_{CO_x}}{N}$$

$$\text{Direct_Assessment_}Score_{CO_x} = \sum_{i=1}^{M} \sum_{j=1}^{N} \frac{Assessment_Score_{CO_x}}{N}$$

Where N is the total number of students enrolled in the course and M are the total types of assessments.

AVERAGE

AVERAGE

	Internal		Mid Sem		End Sem			
	AVG. PERC.	AVG. SCORE	AVG. PERC.	AVG. SCORE	AVG. PERC.	AVG. SCORE	Direct_Assessment_Percentage$_{CO}$ over, all assessments	Direct_Assessment_Score$_{CO}$ over, all assessments
Student 1	50	2	50	2	60	2	**53.33**	2.00
Student 2	70	3	60	2	70	3	66.67	2.67
Student 3	50	2	100	3	50	2	66.67	2.33
Student 4	80	3	100	3	80	3	86.67	3.00
Student 5	80	3	80	3	50	2	70.00	2.67
Student 6	40	1	80	3	90	3	70.00	2.33
Student 7	90	3	70	3	70	3	76.67	3.00
Student 8	80	3	60	2	50	2	63.33	2.33
Student 9	30	1	80	3	60	2	56.67	2.00
Student 10	50	2	50	2	90	3	63.33	2.33
							= 67.33%	**= 2.47**

AVERAGE Over, all students

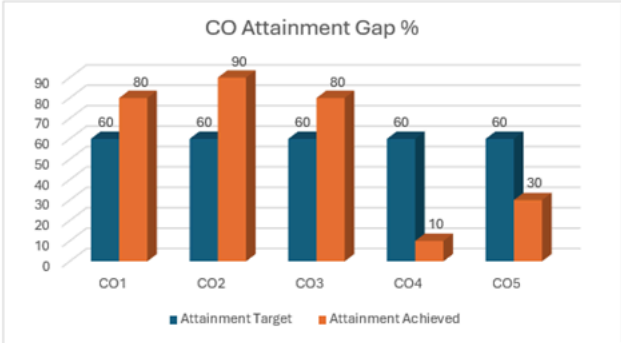
AVERAGE Over, all students

Calculate this for all the other COs, before proceeding to the next Step

4. Direct_Assessment_Percentage$_{CO_x}$ and Direct_Assessment_Score$_{CO_x}$ for all the CO's are tabulated, and an average value is computed (highlighted in Red)

		CO1	CO2	CO3	CO4	CO5	Average
Average Attainment % over all students	**Direct_Assessment_Percentage$_{CO_x}$**	**67.33**	**79.53**	**94.33**	**53.67**	**59.00**	**70.77**
Average Attainment level over all students	**Direct_Assessment_Score$_{CO_x}$**	**2.47**	**2.30**	**2.23**	**1.93**	**1.97**	**2.18**

5. The Direct Attainment of CO's are matched with the pre-defined Targets, to check if the CO's have been attained or not.
TARGET :- The same target value is set for all the COs of a course.
TARGET SET:- 60% STUDENTS MUST ACHIEVE 60% AND ABOVE

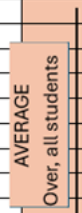


	No. of Students with **Direct_Assessment_Percentage** >60%	% of Students achieving the Target	Attainment Target	Target Achieved?	CO Attainment Gap %	Actions Proposed to Fill the Gap	Modification of CO target (if required)
CO1	8	$\frac{8}{10} * 100 = 80\%$	60%	Yes	+20		
CO2	9	90%	60%	Yes	+30		
CO3	8	80%	60%	Yes	+20		
CO4	1	10%	60%	No	-50		
CO5	3	30%	60%	No	-30		

- If targets are achieved, we may set higher targets subsequently as part of continuous improvement.
- If targets are not achieved then instead of lowering the target, an action plan should be put into place to attain the target in subsequent years.

6. Calculate the indirect assessment score. For this, take the Course Exit feedback from students on all the CO's and compute the average feedback score. The score for each question will be in 0,1,2 or 3 grade.

	CO1	CO2	CO3	CO4	CO5
Student 1	3	3	3	2	1
Student 2	2	3	2	3	3
Student 3	3	2	3	3	2
Student 4	2	3	2	3	2
Student 5	3	3	3	1	2
Student 6	3	3	2	3	3
Student 7	3	3	3	3	3
Student 8	2	2	2	2	2
Student 9	3	3	3	2	1
Student 10	3	3	2	3	1
Indirect_Assessment_$Score_{CO_x}$	**2.7**	**2.8**	**2.5**	**2.5**	**2**



7. Calculate the final CO attainment for the course by taking 70% of Direct Assessment and 30% of the Indirect Assessment

$$Attainment_Score_{CO_x} = 70\%\ \text{Direct_Assessment}_Score_{CO_x} + 30\%\ \text{Indirect_Assessment}_Score_{CO_x}$$

	CO1	CO2	CO3	CO4	CO5	Average
Direct_Assessment_$Score_{CO_x}$- A	2.47	2.30	2.23	1.93	1.97	**2.18**
Indirect_Assessment_$Score_{CO_x}$- B	2.70	2.80	2.50	2.50	2.00	**2.5**
Attainment_Score$_{CO_x}$**=70%*A+30%*B**	**2.54**	**2.45**	**2.31**	**2.10**	**1.98**	**2.01**

Utilized later for PO/PSO attainment

8. Calculate the Direct PO/PSO assessment for all courses in a program, by using the CO-PO/PSO articulation matrix, comprising the level of mapping of a certain CO to a particular PO/PSO. Note, the level of mapping, represented as $Level_{CO-PO_{x,y}}/Level_{CO-PSO_{x,y}}$ in the equation below can be 0, 1, 2 or 3, representing No, Low, Medium and High respectively.
   - Compute the Average of Direct_Assessment_$Score_{CO_x}$ for all CO's, representing the overall Direct_Assessment_Score of the course. Similarly, compute the Average of $Level_{CO-PO}/Level_{CO-PSO}$ for all PO's/PSO's.
   - Populate a table, comprising of Direct_Assessment_$Score_{CO_x}$ for each course in the program and their corresponding average $Level_{CO-PO}/Level_{CO-PSO}$

$$Direct_Assessment_Score_{PO_y} = \frac{\sum_{x=1}^{5} \text{Direct_Assessment}_Score_{CO_x} * \sum_{x=1}^{5} Level_{CO-PO_{x,y}}}{3}$$

$$Direct_Assessment_Score_{PSO_y} = \frac{\sum_{x=1}^{5} \text{Direct_Assessment}_Score_{CO_x} * \sum_{x=1}^{5} Level_{CO-PSO_{x,y}}}{3}$$

where y is the number of POs and PSOs.

The final course outcomes are given in Table 3.7.2.1.

Table 3.7.2.1: The attainments of Course Outcomes.

Sl.No.	Course Code	Course Name	CO1	CO2	CO3	CO4	CO5
1.	MA203	Numerical Methods	2.44	2.505	2.46	2.2	2.22
2.	ME201	Thermodynamics	2.63	2.71	2.54	2.53	2.53
3.	ME203	Fluid Mechanics and Hydraulic Machines	2.16	2.00	2.02	2.41	2.42
4.	ME205	Strength of Materials	2.29	2.015	2.25	2.07	2.215
5.	MA204	Numerical Methods Lab	1.96	1.96	1.46	1.46	1.47
6.	ME202	Fluid Mechanics and Hydraulic Machines Lab	2.24	1.95	2.06	1.97	2.49
7.	ME204	Mechanical Engineering Lab I	2.93	2.88	2.9	2.9	2.9
8.	MT131	Understanding Harmony (UHV2)	2.1	2.1363	2.0007	2.0517	1.5774
9.	ME207	Kinetics and Dynamics of Machines	2.1	2.295	2.4	2.31	2.2

10.	ME209	Energy Conversion Systems	2.35	2.47	2.34	2.34	1.74
11.	ME211	Machine Design	1.45	1.57	0.98	0.98	2.86
12.	ME213	Thermo Fluid Engineering	1.28	1.53	1.19	1.11	2.15
13.	ME208	Dynamics of Machines Lab	2.91	2.81	2.86	2.86	2.86
14.	ME301	IC Engine & Gas Turbine	2.055	2.085	1.705	1.47	1.97
15.	ME303	Mechanical Vibration	2.015	2.025	1.925	2.03	2.065
16.	ME315	Heat and Mass Transfer	1.53	1.53	1.52	1.52	1.57
17.	ME351	Finite Element Methods	2.04	1.96	1.58	1.97	2.39
18.	ME349	Turbo Machinery	0.8	1.2	1.4	0.6	1.8
19.	ME355	Advanced Solid Mechanics	2.16	1.98	2.03	2.17	1.97
20.	ME377	Mechatronics	**1.59**	**1.59**	**1.59**	**1.59**	2.0
21.	ME367	Industrial Tribology	2.38	2.14	2.05	2.4	
22.	ME302	Heat Transfer Lab.	**2.24**	**2.75**	**2.75**	**2.75**	**2.31**
23.	ME304	Internal Combustion Engine Lab	1.95	1.95	1.95	1.51	1.51
24.	ME306	Mechanical Engineering Lab.-II	3	3	3	3	3
25.	ME311	Computer Aided Design	2.525	2.18	2.29	2.065	2.755
26.	ME305	Automobile Engineering	2.265	2.385	2.21	1.83	2.76
27.	ME307	Robotics Engineering	2.26	2.41	2.45	2.52	2.865
28.	ME359	Power Plant Engineering	2.54	2.4	2.4	2.8	2.92
29.	ME369	Gas Dynamics	2.76	2.6	2.75	2.62	2.68
30.	ME373	Hydraulic and Pneumatic Control	2.24	2.21	2.66	2.55	2.55
31.	ME308	Robotics and Automation Lab.	2.67	2.67	2.76	2.76	2.88
32.	ME310	Automobile Engineering Lab.	2.75	2.74	2.72	2.77	2.77
33.	ME401	Refrigeration and Air Conditioning	2.32	1.525	2.69	2.595	2.84
34.	ME409	Industrial Management	2.89	2.89	2.81	2.94	2.96
35.	ME479	Advanced Heat Transfer	1.7	2.25	2.3	2.6	
36.	ME400M	Minor Project	2.9	2.9	2.9	2.9	2.9
37.	ME404	Refrigeration and Air Conditioning Lab	2.49	2.49	2.445	2.7	2.67

38.	ME406	Computer Aided Design and Drafting Lab	2.81	2.87	2.74	2.75	2.75
39.	ME400	Research Project	2.93	2.93	2.93	2.93	2.93

3.8 Attainment of Program Outcomes and Program Specific Outcomes (25)

PO Attainment

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO
MA203	2.37	2.37	2.33	2.33	2.33	2.17	2.16	2.16	2.16	2.29	2.3
ME201	2.60	2.60	2.54	2.54	PO5	PO6	PO7	2.60	2.60	2.60	2.6
ME203	1.70	1.71	1.80	1.83	1.62	PO6	PO7	PO8	2.25	2.25	2.2
ME205	2.17	2.15	2.15	2.16	2.215	2.17	2.17	2.215	2.155	2.07	2.2
MA204	1.63	1.63	1.666	1.66	1.66	1.66	1.47	1.47	1.46	1.59	1.5
ME202	2.13	2.14	2.12	2.16	2.14	PO6	PO7	PO8	2.14	2.14	2.1
ME204	2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9
ME207	2.26	2.26	2.26	2.26	2.26	2.26	2.26	2.26	2.24	2.15	2.2
ME303	2.01	2.01	2.01	2.01	2.01	2.01	2.01	2.01	2.01	2.01	2.0
ME377	1.67	1.67	1.71	1.74	1.68	1.68	1.68	1.69	1.73	1.71	1.8
ME304	1.77	1.73	1.69	1.66	1.73	1.58	1.73	1.66	1.73	1.8	1.7
ME306	3	3	3	3	3	3	3	3	3	3	3
ME311	2.36	2.36	2.39	2.36	2.38	2.36	2.38	2.36	2.36	2.36	2.3
ME305	2.29	2.29	2.29	2.29	2.29	2.28	2.29	2.3	2.3	2.3	2.3
ME359	2.61	2.61	2.61	2.62	2.61	2.61	2.61	2.61	2.61	2.61	2.6
ME369	2.68	2.68	2.69	2.69	2.68	2.69	2.68	2.68	2.68	2.7	2.6
ME308	2.75	2.75	2.75	2.75	2.75	2.75	2.75	PO8	2.75	2.75	PO
ME310	2.75	2.75	2.75	2.75	2.76	2.76	2.76	2.75	2.75	2.75	2.7
ME401	2.3	2.3	2.37	2.36	2.50	2.4	2.39	2.27	2.34	2.40	2.1
ME409	2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9
ME479	2.2	2.2	2.2	2.2	2.25	2.25	2.25	2.2	2.25	2.2	2.3
ME400M	2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9
ME404	2.55	2.54	2.68	2.68	PO5	2.56	2.56	2.56	2.57	2.56	2.5
ME406	2.84	2.84	2.84	2.84	2.84	2.77	2.75	2.75	2.75	2.75	2.7
ME400	2.93	2.93	2.93	2.93	2.93	2.93	2.93	2.93	2.93	2.93	2.9
ME209	2.25	2.25	2.04	2.04	PO5	PO6	PO7	2.25	2.25	PO10	PO
ME211	1.57	1.57	1.63	1.64	1.57	1.57	1.57	1.57	PO9	PO10	1.8
ME213	1.45	1.37	1.45	1.41	1.5	1.43	1.45	1.45	1.45	1.45	PO
MT131	PO1	PO2	2.162	PO4	1.74	2.22	2.13	2.15	1.74	2.04	PO
ME208	2.86	2.86	2.86	2.86	2.86	2.86	2.86	2.86	2.86	2.86	2.8

ME301	1.86	1.81	1.85	1.84	1.76	1.71	1.66	PO8	PO9	1.77	1.7
ME302	2.56	2.56	2.56	2.56	2.56	2.56	2.56	2.56	2.56	2.56	PO
ME307	2.52	2.51	2.52	2.54	2.54	PO6	PO7	PO8	PO9	PO10	PO
ME315	1.53	1.53	1.53	1.52	1.52	1.53	PO7	PO8	PO9	PO10	1.5
ME349	1.16	1.16	1.11	1.13	1.12	PO6	PO7	PO8	1.2	1.2	1.2
ME355	2.06	2.06	2.06	2.05	2.06	2.06	2.06	2.04	2.05	2.16	PO
ME367	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	2.24	PO10	PO
ME373	2.46	2.45	2.45	2.46	2.46	PO6	PO7	PO8	PO9	PO10	PO
ME351	1.99	1.99	1.98	2.03	1.98	1.99	1.98	2.06	2.02	1.98	1.9

PO Attainment Indirect

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO
Exit Feedbac	2.18	2.0	2.33	3	3	2.67	3	3	2.75	2.75	2
Industry Feec	2.8	2.83	2.89	2.89	2.89	2.66	2.66	2.33	2.67	2.67	2.6
Parent Feedt	3	3	3	3	3	3	3	3	3	3	3

PO Attainment Level

> Note: The Institution can fix the weightage of the indirect attainment maximum up to 20%.

Define the Weightage for Indirect Attainment: 20.00

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO
Direct Attainment	2.28	2.28	2.28	2.29	2.26	2.30	2.32	2.33	2.32	2.32	2.
InDirect Attainment	2.66	2.61	2.74	2.96	2.96	2.78	2.89	2.78	2.81	2.81	2.
Overall Attainment	2.36	2.35	2.37	2.42	2.4	2.40	2.43	2.42	2.42	2.42	2.

PSO Attainment

Course	PSO1	PSO2	PSO3
ME201	2.60	2.60	2.60
MA204	1.63	1.66	1.63
ME202	2.14	2.14	2.14
ME204	2.9	2.9	2.9
ME207	2.26	2.23	2.26
ME209	2.25	2.25	2.25
ME211	1.57	1.57	1.57

ME213	1.45	1.47	1.42
ME208	2.86	2.86	2.86
ME301	1.73	1.73	1.73
ME301	1.73	1.73	1.73
ME303	2.03	2.05	2.05
ME351	1.96	1.99	1.99
ME367	2.24	2.24	2.24
ME306	3	3	3
ME304	1.78	1.77	1.77
ME311	2.36	2.36	2.33
ME307	2.47	2.50	2.50
ME373	2.44	2.44	2.44
ME308	2.75	2.75	PSO3
ME310	2.75	2.75	2.75
ME401	2.36	2.39	2.39
ME409	2.9	2.9	2.9
ME479	2.2	2.25	2.2
ME400M	2.9	2.9	2.9
ME404	2.55	2.56	2.56
ME406	2.79	2.78	2.78
ME400	2.93	2.93	PSO3
ME203	2.21	2.17	PSO3
ME205	2.17	2.17	PSO3
ME315	1.53	1.53	PSO3
ME359	2.59	2.61	PSO3
ME369	2.68	2.68	PSO3

PSO Attainment Indirect

Survey	**PSO1**	**PSO2**	**PSO3**
Exit Feedback	2.72	2.72	2.72
Industry Feedback	2.67	2.82	2.67
Parent Feedback	3	3	3

PSO Attainment Level

Course	**PSO1**	**PSO2**

Direct Attainment	2.32	2.33
InDirect Attainment	2.80	2.85
Overall Attainment	2.42	2.43